



Profile summary

107	Total documents
87	Publications indexed in Web of Science
87	Web of Science Core Collection publications
0	Preprints
0	Dissertations or Theses
20	Non-indexed publications
20	Verified peer reviews
0	Verified editor records

Web of Science Core Collection metrics

24	87
H-Index	Publications
1,889	1,692
Sum of Times Cited	Citing Articles
1	1
Sum of Times Cited by Patents	Citing Patents

How My Expertise and Research Plan Will Deliver the EchoHEALTH Postdoctoral Project

2025: Bio-informatics-Area Expert : Instytut Chemii Bioorganicznej Polskiej Akademii Nauk, Poland.
2024-2024: Academic Staff Member, Universitat Potsdam, Germany
2022-2023: Career Addition: Deep Learning, Data Science, Certification.
July 2017- Dec 2021: Postdoctoral Researcher, Finnish Museum of Natural History, Finland.
Aug 2016-Jan 2017: University Fellow, University of Connecticut, Storrs, USA.
Sept 2014-Sept 2016: Research Fellow, University of Technology Sydney, Australia.
Sept 2011-Sept 2014: CoCoPro, Fondazione Edmund Mach, IASMA, Italy.
Dec 2010-Aug 2011: Visiting Post-doc Researcher, Huazhong Agricultural University, China.

What i have been working on: Research Areas

Lead Bioinformatician

1. Genome Assembly and Annotation of Plants and Microbial Genomes
2. Transcriptomics and Transcriptome Evolution
3. Evolutionary and Comparative Genomics
4. Databases and Web-server
5. Metagenomics
6. Pangenomics and Algorithmic implementation

Sequencing Technologies used in Research

1. Illumina - NextSeq, NovaSeq.
2. PacBio and PacBioHiFi

Languages that i use for Research Analysis

– Primarily

1. Python
2. Rust.

– Languages that i used previously

1. C++ and R.

2025: Bio-informatics-Area Expert : Instytut Chemii Bioorganicznej Polskiej Akademii Nauk, Poland.

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Dec 2010-Aug 2011: Visiting Post-doc Researcher, Huazhong Agricultural University, China.

Key abilities

1. Highly collaborative
2. Extremely open-minded
3. I treat everyone equal at work.
4. Positive Attitude
5. Highly skilled in learning new approaches.
6. Result Oriented Approach
7. I always say YES and take and grab chances, job offers and opportunities immediately.

SCIENTIFIC RESEARCH FOCUS

Bioinformatics across all species (plants, bacterial, fungal, humans) using Illumina and LongRead sequencing technologies.
My focus: **Microbiome ML, General & Deep Learning in Genomics** (not single-cell or image ML). I analyze single-cell data but do not do ML on it.



RUST

Primary Language for Software

Software Engineering • Performance • Reliability • Safety



Scientific Software



CLI Tools & Applications



Pipelines & Workflows



Web & Services



HPC & High Performance



AI/ML Software & Integrations

I build complete, compiled, production-grade scientific software, pipelines and tools in Rust.



Results
Reproducibility
Scientific
Impact



PYTHON

For Machine Learning & Scientific ML

ML Ecosystem • Productivity • Research • Analysis



Classical ML



Deep Learning



Graph Learning



Feature Eng. & Analysis



Model Eval. & Visualization



Scientific Libraries

I use Python specifically for ML, deep learning and scientific analysis of structured biological, chemical and molecular data.

DATA & RESEARCH DOMAINS (Structured Scientific Data)



DNA / Genomics

Sequence analysis
Genomic prediction
Pangenomes



Microbiome

Microbial ML
Metagenomics
Community analysis



Gene Networks & Interactions

Graph construction
GCN, GAT, GraphSAGE
Network analysis



Metabolomics

Classification
Feature selection
Clustering



Chemical / Molecular Data

Molecular descriptors
Property prediction
QSAR / QSPR



Biological Time Series

Time-series analysis
Similarity search
Matrix profiles



Other Structured Data

Tabular data
Multi-omics
Mixed data types

MACHINE LEARNING & ANALYTICAL TOOLKIT (Python Ecosystem)

Classical ML (scikit-learn)

- SVM
- Random Forest
- Gradient Boosting
- DBSCAN
- k-Means
- PCA / t-SNE
- Pipelines & CV

Gradient Boosting (LightGBM / XGBoost / CatBoost)

- LightGBM
- XGBoost
- CatBoost
- Feature Importance
- Hyperparameter Tuning

Deep Learning (PyTorch / fastai)

- CNN / Conv1D
- Autoencoders
- Transformers
- Sequence Models
- Transfer Learning
- fastai high-level API

Graph ML (PyG / DGL)

- GCN
- GAT
- GraphSAGE
- Link Prediction
- Node Prediction
- Graph Classification

Time Series (STUMPY / sktime)

- Matrix Profile
- Motif Discovery
- Anomaly Detection
- Similarity Search
- Forecasting

Scientific Libraries & Ecosystem

- NumPy / SciPy
- pandas / Polars
- Biopython
- RDKit
- Scanpy
- Matplotlib / Seaborn



ROLE OF R



Statistical Analysis



Bioinformatics Analysis



Statistical Visualization



Specialized Bioconductor Tools



Used where R is the most appropriate tool

TYPICAL RESEARCH WORKFLOW



Data Collection



Preprocessing & QC



Feature Engineering



Modeling (ML/DL)



Evaluation & Validation



Biological Interpretation



Scientific Software & Reproducible Pipelines (Rust)

TECHNOLOGY PHILOSOPHY

- ✓ Results over language. I use the best tool for the scientific problem.
- ✓ Rust for building reliable, high-performance scientific software.
- ✓ Python for cutting-edge ML and scientific analysis.
- ✓ R for statistics and analyses where it excels.
- ✓ Reproducibility, correctness and scientific impact are paramount.

KEY PRINCIPLES



Scientific Research First



Structured Data ML Focus



Not Image ML / Not Single-cell ML



Quality, Reproducibility, Impact



Collaboration within employer framework only



A Modern Stack for Bioinformatics & Computational Biology

Python • Rust • Bash — A powerful combination for biology at scale



PYTHON — PRIMARY

Analysis • ML • Data Science



Bioinformatics

Analyze sequences, structures, annotations and more.



Computational Biology

Model biological systems and simulate complex processes.



Machine Learning / AI

Build predictive models for genomics, proteomics and other omics.



Scientific Computing

Numerical methods, statistics, visualization and simulation.



Data Processing

Clean, transform and analyze large biological datasets.



APIs & Automation

Build APIs, automate tasks and integrate tools and services.

```
import pandas as pd
from Bio import SeqIO
from sklearn.ensemble import
    RandomForestClassifier

# Load sequence data
records = list(SeqIO.parse(
    "sequences.fasta", "fasta"))
data = []
for r in records:
    data.append(len(r.seq))

df = pd.DataFrame({'len': data})

# Example ML model
X = df[['len']]
y = (df['len'] > 1000).astype(int)

model = RandomForestClassifier()
model.fit(X, y)

print("Model trained 🍀")
```

Key Libraries



Biopython



NumPy



pandas



Learn



Matplotlib



Seaborn



RUST — PERFORMANCE

High Performance • Safe • Concurrent



High-Performance Bioinformatics

Fast, memory-safe tools for compute-intensive tasks.



Sequence Processing

Efficient parsing, alignment and manipulation.



Genomics Tools

Build reliable tools for genomics and large-scale analysis.



CLI Applications

Fast, portable and user-friendly CLI tools.



Avoid Computational Bottlenecks

Remplement hotspots and accelerate pipelines.

```
use bio::io::fasta;
use bio::stats::gc_content;

fn main() -> Result<(), Box<dyn
std::error::Error>> {

    let reader = fasta::Reader::from_file(
        ("in.fasta"));

    for record in reader.records() {
        let rec = record?;
        let seq = rec.seq();
        let gc = gc_content(seq);
        println!("{}", rec.id(), gc);
    }

    Ok(())
}
```

Why Rust?



Blazing fast



Memory safe



Reliable



Perfect for systems work



BASH — INFRASTRUCTURE / GLUE

Automation • Pipelines • Orchestration



Linux & HPC

Work in the environment where bioinformatics lives.



Bioinformatics Pipelines

Chain tools together seamlessly.



Scientific Tools

Integrate CLI tools and software.



Large Data Handling

Handle huge files and stream data efficiently.



Workflow Automation

Reproducible, repeatable and scalable pipelines.

```
#!/usr/bin/env bash
set -euo pipefail

IN_FASTQ="data/sample.fq.gz"
OUT_DIR="results"

mkdir -p "$OUT_DIR"

# 1. Quality control
fastqc "$IN_FASTQ" -o "$OUT_DIR/qc"

# 2. Trim adapters
trim_galore "$IN_FASTQ" -o "$OUT_DIR/trimmed"

# 3. Align to reference
bwa mem ref.fa "$OUT_DIR/trimmed/
"${basename "$IN_FASTQ"}.fq.gz" |
samtools sort -o "$OUT_DIR/aligned.bam"

# 4. Count reads
featureCounts -a genes.gtf -o "$OUT_DIR/
counts.txt" "$OUT_DIR/aligned.bam"

echo "Pipeline completed 🍀"
```

Essential Commands



grep



awk



sed



xargs



find

HOW IT ALL WORKS TOGETHER

1. Biological Question

Define the question and gather data.

2. Bash Pipeline

Download data, run tools, clean and prepare.

3. Rust Processing

High-performance processing, parsing, alignment, counting.

4. Python Analysis

Analyze results, ML/AI, statistics, visualization.

5. Biological Insight

Interpret results and generate new hypotheses.



REPRODUCIBLE

Automated, version-controlled and repeatable workflows.



SCALABLE

Handle small experiments to massive datasets.



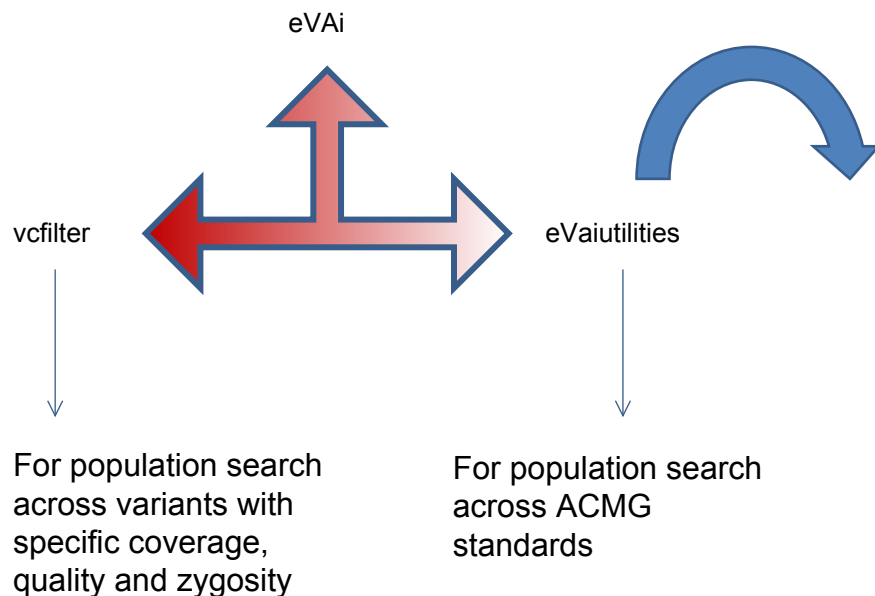
RELIABLE

Robust tools, low-level performance, high-level productivity.



DRIVEN BY BIOLOGY

Technology serves discovery and advances life science.



NAR Molecular Medicine, 2026, 3, ugag023
<https://doi.org/10.1093/narmme/ugag023>
 Advance access publication date: 30 April 2026
 Repeat Expansion Diseases

OXFORD

CAG-targeting artificial miRNA with reduced off-target risk for efficient lowering of pathogenic polyglutamine proteins

Marianna Pewinska-Kolodziejczak¹, Anna Kotowska-Zimmer¹, Lukasz Przybyl², Dorota Wronka², Anna Karlik², Michal Smuszkievicz¹, Julia Balcerek¹, Joanna Suszynska-Zajczyk³, Anna Piaszyk-Borychowska¹, Martyna Urbanek-Trzeciak⁴, Gaurav Sablok⁵, Emilia Kozłowska⁶, Jan Podkowinski⁵, Agnieszka Fisz⁶, Piotr Kozłowski⁴, Luiza Handschuh⁵, Marta Olejniczak^{1,*}

¹Department of Genome Engineering, Institute of Bioorganic Chemistry, Polish Academy of Sciences, Noskowskiego 12/14, 61-704 Poznan, Poland

²Laboratory of Mammalian Model Organisms, Institute of Bioorganic Chemistry, Polish Academy of Sciences, Noskowskiego 12/14, 61-704 Poznan, Poland

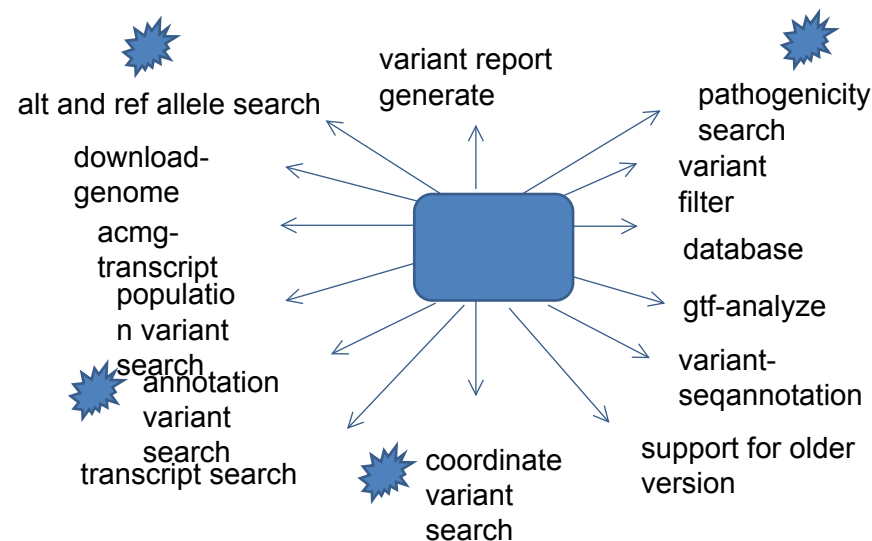
³Department of Biochemistry and Biotechnology, Poznan University of Life Sciences, 60-632 Poznan, Poland

⁴Department of Molecular Genetics, Institute of Bioorganic Chemistry, Polish Academy of Sciences, Noskowskiego 12/14, 61-704 Poznan, Poland

⁵Laboratory of Genomics, Institute of Bioorganic Chemistry, Polish Academy of Sciences, Noskowskiego 12/14, 61-704 Poznan, Poland

⁶Department of Medical Biotechnology, Institute of Bioorganic Chemistry, Polish Academy of Sciences, Noskowskiego 12/14, 61-704 Poznan, Poland

*To whom correspondence should be addressed. Email: marta.olejniczak@ibch.poznan.pl



Instytut Chemii Bioorganicznej Polskiej Akademii Nauk:

1. Development of softwares for human genome analysis.
2. Bioinformatics data analysis related to RNA-Seq: Genome mapping, Differential analysis, transcript abundance analysis.
3. Completed 3 projects with the analysis and developed Rust based human genome analysis software.

evaiweb

Search evai annotation output by any field below

GENE REFERENCE (GENEREf) e.g. NM_000000 <button>Search</button>	START e.g. 12345 <button>Search</button>	STOP e.g. 12400 <button>Search</button>	ALT e.g. A <button>Search</button>
GENE e.g. TP53 <button>Search</button>	GENOTYPE e.g. het <button>Search</button>	PATHOGENICITY SCORE e.g. 0.98 <button>Search</button>	

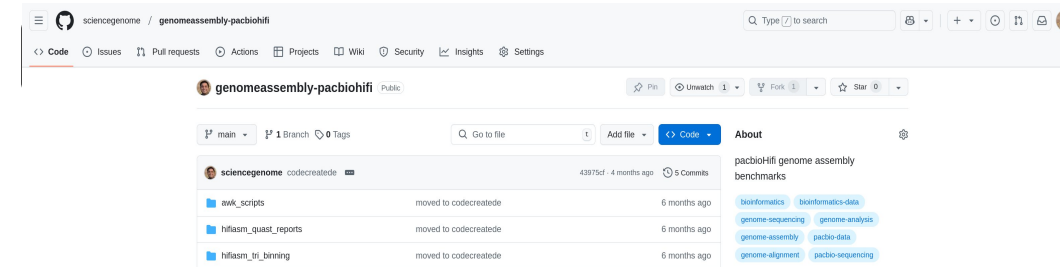
<https://codeberg.org/gsablock/evaiweb>

Matches for "AG"

172 result(s) found

[New search](#)

Gene ref	Sample	Chrom	Start	Stop	Alt	HGVSp	Gene	Final class
AG	CLL_2081_3232	1	788419	788420	-	null	AL669831.5	Benign
AG	CLL_2081_3232	1	788419	788420	-	null	AL669831.3	Benign
AG	CLL_2065_2423	1	788419	788420	-	null	AL669831.5	Benign
AG	CLL_2065_2423	1	788419	788420	-	null	AL669831.3	Benign
AG	CLL_2070_2477	1	788419	788420	-	null	AL669831.5	Benign
AG	CLL_2070_2477	1	788419	788420	-	null	AL669831.3	Benign
AG	CLL_2055_1883	1	788419	788420	-	null	AL669831.5	Benign
AG	CLL_2055_1883	1	788419	788420	-	null	AL669831.3	Benign
AG	CLL_2043_1222	1	788419	788420	-	null	AL669831.5	Benign



genomeassembly-pacbiohifi

I completed analysis and benchmarking of the pacbiohifi for the genome assembly coming from the PacBioHifi reads for the genotypes and haplotypes and their quality assessment. This is the complete genome assembly standardization coming from the PacBioHifi reads and includes all the scripts and the analysis for the reproducibility and the visualization and the summary analysis.

Genome assembly packages used for the assembly

- [hifiasm](#)
- [Meryl](#)
- [Merqury](#)
- [Verkko](#)

2024-2024: Academic Staff Member, Universitat Potsdam, Germany

Objective:

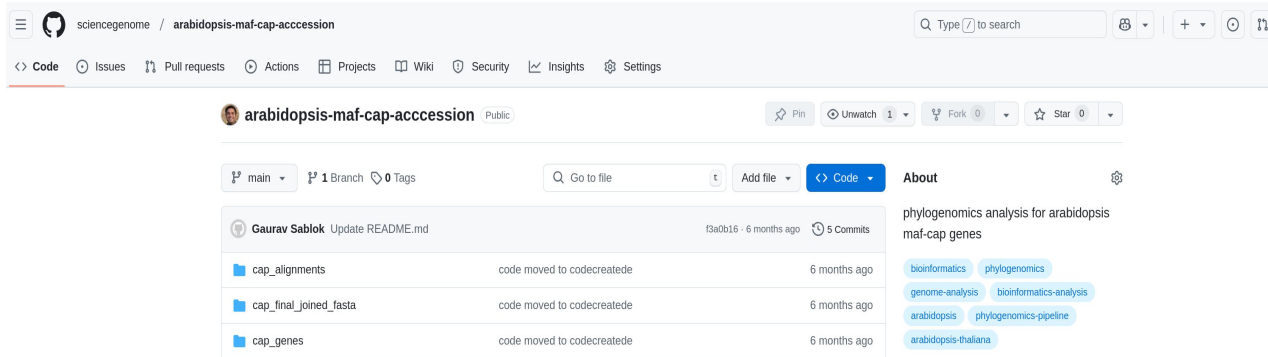
1. Benchmark all the tools for the genome pacbiohifi.
2. Establish the best suitable approaches.
3. A thorough and comparative analysis.
4. A web based information system for the same.

Assembly of pangenomes

<https://codeberg.org/gsablok/longpan>

Annotation and Graph pangenomes

<https://codeberg.org/gsablok/graphpan>



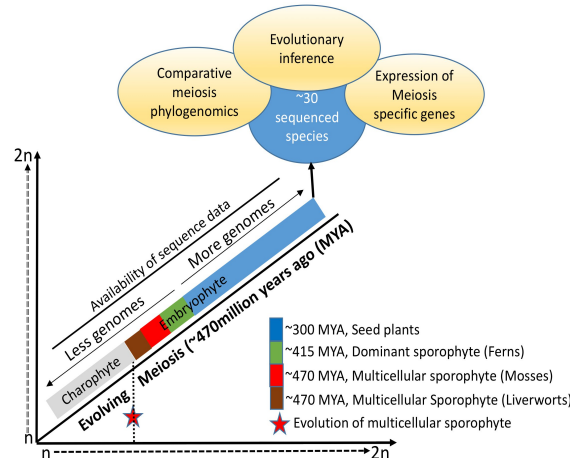
Objective:

1. Analyze the *Arabidopsis thaliana* maf cap genes across all *Arabidopsis* genomes and phenotypes.
2. Phylogenomics analysis and genome analysis and establishing collinearity analysis.

July 2017- Dec 2021: Postdoctoral Researcher, Finnish Museum of Natural History, Finland.

1. Genome of *Blasia*, *Coleochaete*, *Chaetosporidium*, *Polytrichum* genome
2. Chloroplast genome of *Blasia*, *Coleochaete*, *Chaetosporidium*, *Polytrichum* genome
3. Mitochondrial genome of *Blasia*, *Coleochaete*, *Chaetosporidium*, *Polytrichum* genome
4. *Blasia* transcriptome
5. Genome of *Cryptomonas* and *Mallomonas*
6. Chloroplast genome of *Cryptomonas* and *Mallomonas*
7. Light assisted transcriptomics of *Cryptomonas* and *Mallomonas*.
8. Chloroplast genome analysis of *Marattiaceae* and *Ambrosia* species.
10. Chloroplast genome analysis of an established collaboration with Edinburgh, UK.
11. Genome assembly, annotation, genome completeness, marker genes identification, ITS identification, phylogenomics of more than 500 fungal genomes sequenced at Finnish Museum of natural History, Helsinki Finland.

Lead developer using C++ till 2021, doing the extractions, genome assembly, writing algorithms, evolutionary and comparative genomics, server and cluster management including cloudops.



LUOMUS



5th Conference on Plant Genome Evolution
29 September – 1 October 2019 • Sitges, Spain

ACADEMY OF FINLAND

First draft genome assembly of *Coleochaete orbicularis* for genome scale comparisons across embryophytes

Gaurav Sablok^{1,2}, Xiaolan He¹, Mari Miranto¹, Elina Peltomaa³, Robin Sleith⁴, Kenneth Karol⁵, Charles Delwiche⁶, Neil Bell⁶, Lars Paulin⁷, Peter Pocai⁸, Jaakko Hyvönen^{1,2*}

¹Finnish Museum of Natural History, University of Helsinki, Finland; ²OEB, Faculty of Biology and Environmental Sciences, University of Helsinki, Finland; ³ECOEVO, Faculty of Biology and Environmental Sciences, University of Helsinki, Finland; ⁴New York Botanical Garden, USA; ⁵University of Maryland USA; ⁶Royal Botanic Garden, Edinburgh, UK; ⁷Institute of Biotechnology, University of Helsinki, Finland.

LUOMUS



3rd Agriculture and Climate Change CONFERENCE
Budapest, Hungary | 24 - 26 March 2019

ACADEMY OF FINLAND

Mitogenomics of *Blasia pusilla* as a tool for agricultural productivity

Gaurav Sablok^{1,2}, Xiaolan He¹, Mari Miranto¹, Elina Peltomaa³, Robin Sleith⁴, Kenneth Karol⁵, Charles Delwiche⁶, Neil Bell⁶, Lars Paulin⁷, Peter Pocai⁸, Jaakko Hyvönen^{1,2*}

¹Finnish Museum of Natural History, Helsinki, Finland; ²Organismal and Evolutionary Biology (OEB) Research programme, University of Helsinki, Finland; ³Ecosystems and Environment Research programme (EEG), University of Helsinki, Finland; ⁴New York Botanical Garden, USA; ⁵University of Maryland, USA; ⁶Royal Botanic Garden, Edinburgh, UK; ⁷Institute of Biotechnology, University of Helsinki, Finland.

LUOMUS



ISFMS 2019 3rd International Symposium on Frontiers in Molecular Science
RNA Regulatory Networks
26 – 28 June 2019, Lisbon, Portugal

ACADEMY OF FINLAND

Gametophytic transcriptomics of an early embryophyte (*Blasia pusilla*) and comparative landscape of gametophyte evolution

Gaurav Sablok^{1,2}, Xiaolan He¹, Mari Miranto¹, Elina Peltomaa³, Robin Sleith⁴, Kenneth Karol⁵, Charles Delwiche⁶, Neil Bell⁶, Lars Paulin⁷, Peter Pocai⁸, Jaakko Hyvönen^{1,2*}

¹Finnish Museum of Natural History, University of Helsinki, Finland; ²OEB, Faculty of Biology and Environmental Sciences, University of Helsinki, Finland; ³ECOEVO, Faculty of Biology and Environmental Science, University of Helsinki, Finland; ⁴New York Botanical Garden, USA; ⁵University of Maryland USA; ⁶Royal Botanic Garden, Edinburgh, UK; ⁷Institute of Biotechnology, University of Helsinki, Finland.

LUOMUS



EMBO Plant EVO 2018

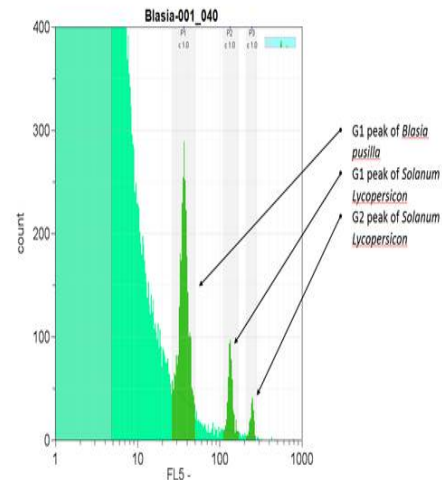
ACADEMY OF FINLAND

The first draft genome of *Blasia pusilla* using PACBIO-SMRT sequencing

Gaurav Sablok^{1,2}, Xiaolan He¹, Mari Miranto¹, Elina Peltomaa³, Neil Bell⁴, Lars Paulin⁵, Peter Pocai¹, Jaakko Hyvönen^{1,2*}

¹Finnish Museum of Natural History, Helsinki, Finland; ²OEB, Faculty of Biological and Environmental Science, University of Helsinki, Finland; ³ECOEVO, Faculty of Biological and Environmental Science, University of Helsinki, Finland; ⁴Royal Botanic Garden, Edinburgh, UK; ⁵Institute of Biotechnology, Helsinki, Finland

*Email: jaakko.hyvonen@helsinki.fi

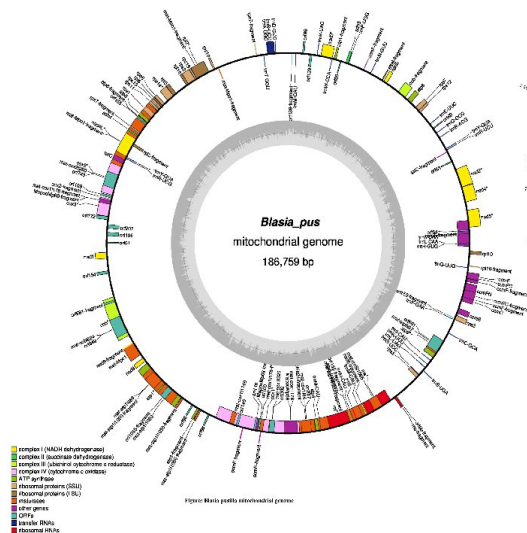


Blasia pusilla

Transcriptome Assembled transcripts	130871
Number of Supertranscripts	80880
Percent GC	46.55
*Transcriptome statistics based on ALL transcript contigs	
Contig N50:	3191bp
Median contig length	653bp
Average contig	1533.29bp
*Transcriptome statistics based on longest isoform	
Contig N50	2334bp
Median contig length	395bp
Average contig	921.32bp
Number of coding regions predicted using GeneMark-ST (Trau)	60812
Number of coding regions predicted using GeneMark-ST (Sup)	24336
Number of transcripts associated virNOG:	39499
Number of supertranscripts associated virNOG:	11696
Number of transcripts mapped to KEGG:	22996
Number of supertranscripts mapped to KEGG:	7199
Domain assessment of transcriptome completeness: 59.28% completeness representing 77.25% (CDAsize 1), 55. BUSCO assessment of transcriptome completeness: 44% completeness representing 635 (CB), 610 (CSB)	

Blasia pusilla gametophyte transcriptome summary statistics

Genome assembly	Metric
(*Canu=corMhapSensitivity=high, corMinCoverage=0, corOutCoverage=100)	
Total Sequenced PacBio Reads	1,997,826
Organelle and Symbiotic reads	215,908
Reads used for Canu assembly (35.76X coverage) > 1000bp	1,781,918
Contigs (Canu, MHAP overlapper)	4,503
SSPACE Scaffolds (Canu, MHAP overlapper)	2,765
Contigs (Canu, MHAP overlapper + -correctedErrorRate=0.060)	4,834
SSPACE Scaffolds (Canu, MHAP overlapper + -correctedErrorRate=0.060)	2,992
Contigs (Canu-MECAT, 30X)	3,582
SSPACE Scaffolds (Canu-MECAT, 30X)	2,131
Contigs (MARVEL)	12,757
SSPACE Scaffolds (MARVEL)	7,280
N50 Scaffolds (Canu, MHAP overlapper)	242,218bp
N50 Scaffolds (Canu, MHAP overlapper + -correctedErrorRate=0.060)	193,193bp
N50 Scaffolds (Canu-MECAT)	175,995bp
N50 Scaffolds (MARVEL)	117,508bp
Genome Assembly (Canu, MHAP overlapper)	180MB
Genome Assembly (Canu, MHAP overlapper + -correctedErrorRate=0.060)	215MB
Genome Assembly (Canu MECAT, 30X)	206MB
Genome Assembly (MARVEL)	453MB



Yue et al. *Genome Biology* (2020) 27:166
<https://doi.org/10.1186/s13059-020-04101-2>

Genome Biology

RESEARCH

Open Access

Genome assemblies of the liverwort *Blasia pusilla* uncover a well-defined pseudoautosomal region on homomorphic UV sex chromosomes

Yuling Yue^{1,2}, Gaurav Sablok³, Xiaolan He³, Jaakko Hyvönen³, Shanshan Dong⁴, Giacomo Potente^{1,2,4,5}, Yang Liu⁴ and Péter Szövényi^{1,2*}

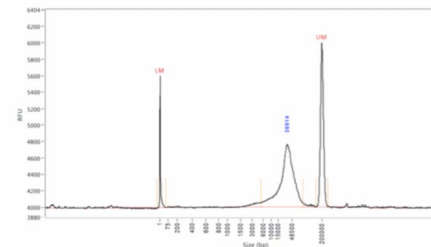
*Correspondence: peter.szovenyi@uzh.ch

¹Department of Systematic and Evolutionary Botany, University of Zurich, Zollikonerstrasse 107, Zurich 8008,

Abstract

Background: Sex chromosomes evolve from autosomes and are expected to be initially homomorphic, though they may become heteromorphic over time. In diploid systems, one sex chromosome often degenerates, whereas homomorphy should per-

Fragment Analyzer

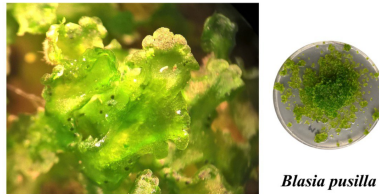


The first draft genome of *Blasia pusilla* using PACBIO-SMRT sequencing

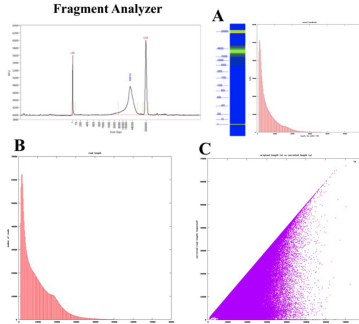
Gaurav Sablok^{1,2}, Xiaolan He¹, Mari Miranto¹, Elina Peltomaa³, Neil Bell⁴, Lars Paulin⁵, Peter Poczai¹, Jaakko Hyvönen^{1,2*}
¹Finnish Museum of Natural History, Helsinki, Finland; ²OEB, Faculty of Biological and Environmental Science, University of Helsinki, Finland; ³ECOEVO, Faculty of Biological and Environmental Science, University of Helsinki, Finland; ⁴Royal Botanic Garden, Edinburgh, UK; ⁵Institute of Biotechnology, Helsinki, Finland
 *Email: jaakko.hyvonen@helsinki.fi

Introduction:

In embryophyte genomics, much of the focus has been on understanding genetic adaptation and molecular evolution of angiosperms. The liverwort *Blasia pusilla*, representing one of the basal clades of embryophytes, has been studied to identify the genetic basis of traits that potentially led to the success of the embryophytes. Comparative genomic analyses will help us understand the key transitions of embryophyte evolution, such as terrestrialization resulting in various innovations driven by genetic change.



Blasia pusilla



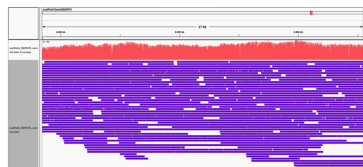
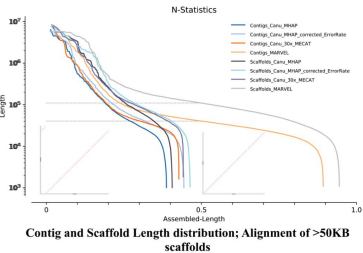
PacBio Read length distribution: A) Read length; B) Corrected Read Length; C) Overlap of read length and corrected read length

Material and Methods:

In-vitro grown gametophytic cultures of *Blasia pusilla* on MS medium were used for genomic DNA extraction using the Plant-EZ DNA extraction kit yielding a total of 25.24 ng gDNA with fragment size ranging between 27-37 KB and were sequenced using PacBio-SMRT at the Institute of Biotechnology, University of Helsinki, Finland. Reads were mapped using BLASR (Chaisson and Tesler 2011) to organelle and cyanobacterial genomes. Filtered reads were assembled using Canu, which is an adaptive k-mer weighting assembler (Soren et al. 2017), MECAT (Xiao et al. 2017) and MARVEL (Nowoshilow et al. 2018) and scaffolded using SSPACE (Boetzer, 2014).

Results: The comparative details of the draft genome assembly are given below:

Genome assembly (*Canu=corMapSensitivity=high, corMinCoverage=0, corOutCoverage=100)	Metric
Total Sequenced PacBio Reads	1,997,826
Organelle and Symbiotic reads	215,908
Reads used for Canu assembly (35.76X coverage) > 1000bp	1,781,918
Contigs (Canu, MHAP overlap)	4,503
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Contigs (Canu-MECAT, 30X)	3,582
SSPACE Scaffolds (Canu-MECAT, 30X)	2,131
Contigs (MARVEL)	12,757
SSPACE Scaffolds (MARVEL)	7,280
N50 Scaffolds (Canu, MHAP overlap)	242,218bp
N50 Scaffolds (Canu, MHAP overlap + -correctedErrorRate(0.060))	193,193bp
N50 Scaffolds (Canu-MECAT)	175,995bp
N50 Scaffolds (MARVEL)	117,508bp
Genome Assembly (Canu, MHAP overlap)	180MB
Genome Assembly (Canu, MHAP overlap + -correctedErrorRate(0.060))	215MB
Genome Assembly (Canu MECAT, 30X)	206MB
Genome Assembly (MARVEL)	453MB



References:
 Boetzer M, Pirovano W (2014) BMC Bioinformatics. 15:211; Chaisson MJ, Tesler G (2012) BMC Bioinformatics. 13:238; Koren et al. (2017) Genome Res. 27:722-736; Nowoshilow et al. (2018) Nature 554:50-55; Xiao et al. (2017) Nat Methods 11:1072-1074

Acknowledgements: Academy of Finland for their grant 295959 to JH & XH, and to CSC - IT Centre for Science for providing computing environment.

Plastomics of charophycean *Coleochaete orbicularis* and embryophyte *Blasia pusilla*

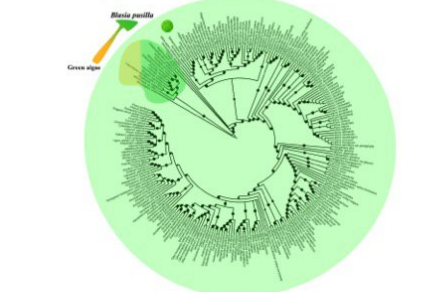
Gaurav Sablok^{1,2}, Xiaolan He¹, Mari Miranto¹, Elina Peltomaa³, Robin Sleith⁴, Kenneth Karol⁵, Charles Delwiche⁶, Neil Bell⁴, Lars Paulin⁵, Peter Poczai¹, Jaakko Hyvönen^{1,2*}
¹Finnish Museum of Natural History, Helsinki, Finland; ²OEB, Faculty of Biological and Environmental Sciences, University of Helsinki, Finland; ³ECOEVO, Faculty of Biological and Environmental Sciences, University of Helsinki, Finland; ⁴New York Botanical Garden, USA; ⁵University of Maryland USA; ⁶Royal Botanic Garden, Edinburgh, UK; ⁷Institute of Biotechnology, University of Helsinki, Finland.
 *Email: jaakko.hyvonen@helsinki.fi

Introduction:

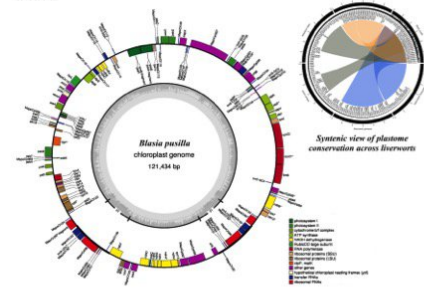
Plastomes play an important role in maintaining plant viability through dynamic remodeling of electron transport across the thylakoid membrane. With few exceptions, embryophyte (land plant) plastomes are structurally conserved, while those of green algae show more variations in gene order and content. These features provide clues towards understanding not only plastome evolution but also the phylogenetic relationships of the plastid lineage. We report the first plastomes of the embryophyte *Blasia pusilla*, representing one of the early lineages of liverworts, and the charophycean green alga *Coleochaete orbicularis*, an epiphyte of shallow fresh water habitats, based on a combination of PACBIO-SMRT and Illumina sequencing technologies. We also provide an overview of the transcriptional status of the *C. orbicularis* plastome and phylogenetic resolution using the embryophyte *B. pusilla*.

Material and Methods:

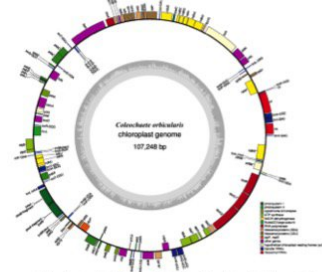
In-vitro grown gametophytic cultures of *B. pusilla* on MS medium were used for genomic DNA extraction with the Plant-EZ DNA extraction kit and were sequenced using PacBio-SMRT at the Institute of Biotechnology, University of Helsinki, Finland. Additionally, Illumina sequencing was performed using Next-Seq with an average insert size of ~450-600bp. In the case of *C. orbicularis* UTEX LB 2651, an Illumina PE library with an average insert size of ~150bp was sequenced using HiSeq2500. Chloroplast reads were extracted by mapping to chloroplast genomes available from NCBI using bowtie2 (Langmead, 2012). Reads were assembled using SPADEx as an assembler and scaffolds were aligned to *Marchantia paleacea* and *Coleochaete scutata* (for *B. pusilla* and *C. orbicularis* respectively) using MAFFT (Katoh et al. 2002). Chloroplast reads were assembled using Canu (Koren et al. 2017) and SPADEx-hybrid (Antipov et al. 2016). Assembly contiguity was evaluated using the LASTZ alignment program (Harris, 2007). Plastome annotations were undertaken using GENE (Tillich et al. 2017) and coverage plots were visualized with Circos (Krzywinski et al. 2009).



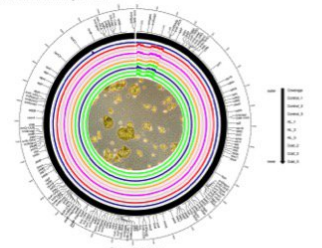
Phylogenetic positioning of *Blasia pusilla* using the concatenated matrix alignment of the *rbcL* gene representing sampling across multiple algal and embryophyte clades. Phylogenetic reconstruction was done using MAFFT (Katoh et al. 2002) and IQ-TREE (Tillich et al. 2016) and visualization with iTOL (Letunic et al. 2016). Bootstrap values are shown for branches supported at 50% or over.



Plastome map of *Blasia pusilla* representing a total of 27,017 PACBIO-SMRT reads with overlap (1856,734X coverage and an N50 of 158,613). Post-assembly correction and trimming represents 615,725 coverage supplemented with additional 631,594 PE-Next-Seq Illumina reads.

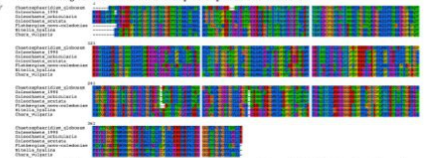


Plastome map of *Coleochaete orbicularis* using 4,822,514 paired end reads yielding a total assembly of 197,248 bp with 99.96% re-mapping rate of the plastomic reads. *C. orbicularis* lacks the inverted repeats, in common with *C. scutata* plastome.



Transcriptome and genome-wide coverage map of *C. orbicularis* representing mapping of 4,872,592 paired-end reads along with colored ribbons displaying the transcriptionally active bases. Transfer RNAs showed higher expression as compared to other genes, which is an evolutionary conserved phenomenon across green plants suggesting up-regulation of translation for combating stress, a mechanism in common with *C. scutata* plastome.

Protein alignment of chloroplast elongation factor Tu (*tufA*) genes across *Coleochaete orbicularis*, *Coleochaete scutata*, *Charophytum globosum* and *Chlamydomonas reinhardtii* was done using MUSCLE (Edgar, 2004) and were visualized using SE-View available from <http://cibioinformatics.org/se-view>



Coleochaete_1990 represents the first record of the chloroplast *tufA* gene following its transfer to the nucleus in green algae. Baldauf et al. PNAS (1996).

References:
 Langmead et al. (2012) Nature Methods; Katoh et al. (2002) Genome Biology; Boetzer et al. (2012) Genome Biology; Koren et al. (2017) Genome Research; Antipov et al. (2016) Bioinformatics; Tillich et al. (2017) Nucleic Acid Research; Krzywinski et al. (2009) PLoS ONE; Tillich et al. (2016) Nucleic Acid Research; Letunic et al. (2016) Nucleic Acid Research; Harris et al. (2007) Genome Research; Harris (2007) PLoS ONE; Letunic et al. (2016) Nucleic Acid Research; Krzywinski et al. (2009) PLoS ONE; Tillich et al. (2016) Nucleic Acid Research.

Acknowledgements: Academy of Finland for their grant 295959 to JH & XH, and to CSC - IT Centre for Science for providing computing environment.

Finnish Museum of Natural History, Helsinki, Finland; Organismal and Evolutionary Biology (OEB) Research programme, University of Helsinki, Finland; Ecosystems and Environment Research programme (EEGS), University of Helsinki, Finland; NewYork Botanical Garden, USA; University of Maryland, USA; Royal Botanic Garden, Edinburgh, UK; Institute of Biotechnology, University of Helsinki, Finland.

Phylogenetic relationships among the 10 strains and model for evolution. Note the ancestral state of the pool of sequences is represented by a yellow-filled triangle. The model for evolution is based on the 10 strains and the 10 strains are represented by a yellow-filled triangle. The model for evolution is based on the 10 strains and the 10 strains are represented by a yellow-filled triangle.

The phylogenetic tree on the left shows the relationships between 10 strains. The strains are: 10, 9, 8, 7, 6, 5, 4, 3, 2, 1. The model for evolution on the right shows the relationships between 10 strains and a model for evolution. The model for evolution is based on the 10 strains and the 10 strains are represented by a yellow-filled triangle.

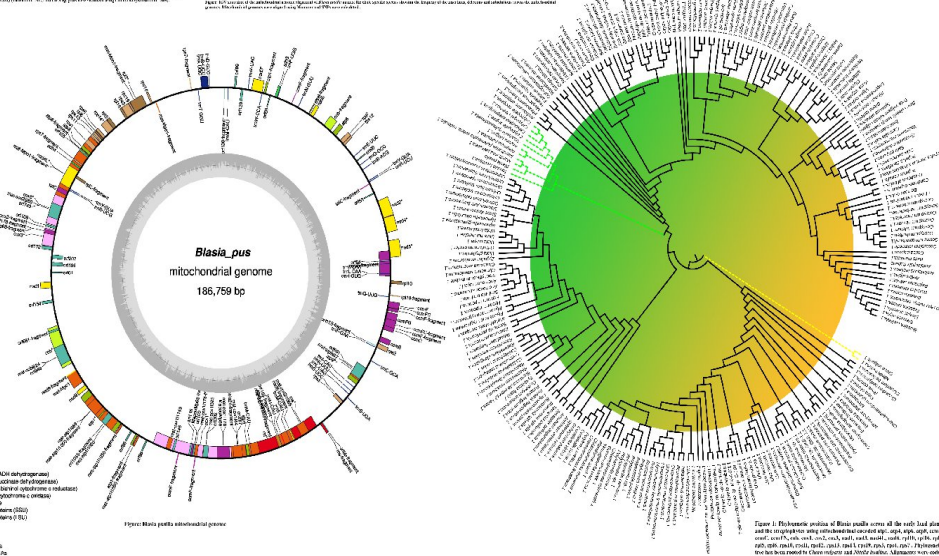
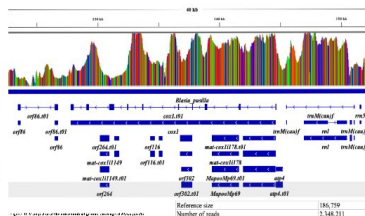
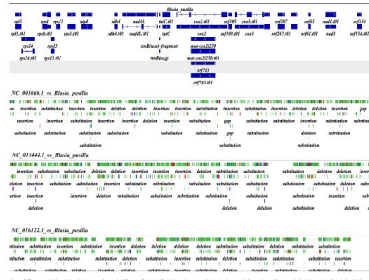
[illegible]

Figure 1. Phylogenetic position of *Blairstipula* across all the nucleotide sites and the microsatellite using intraspecific data (colored) and *ap1*, *ap4*, *ap5*, *ap6*, *ap7*, *ap8*, *ap9*, *ap10*, *ap11*, *ap12*, *ap13*, *ap14*, *ap15*, *ap16*, *ap17*, *ap18*, *ap19*, *ap20*, *ap21*, *ap22*, *ap23*, *ap24*, *ap25*, *ap26*, *ap27*, *ap28*, *ap29*, *ap30*, *ap31*, *ap32*, *ap33*, *ap34*, *ap35*, *ap36*, *ap37*, *ap38*, *ap39*, *ap40*, *ap41*, *ap42*, *ap43*, *ap44*, *ap45*, *ap46*, *ap47*, *ap48*, *ap49*, *ap50*, *ap51*, *ap52*, *ap53*, *ap54*, *ap55*, *ap56*, *ap57*, *ap58*, *ap59*, *ap60*, *ap61*, *ap62*, *ap63*, *ap64*, *ap65*, *ap66*, *ap67*, *ap68*, *ap69*, *ap70*, *ap71*, *ap72*, *ap73*, *ap74*, *ap75*, *ap76*, *ap77*, *ap78*, *ap79*, *ap80*, *ap81*, *ap82*, *ap83*, *ap84*, *ap85*, *ap86*, *ap87*, *ap88*, *ap89*, *ap90*, *ap91*, *ap92*, *ap93*, *ap94*, *ap95*, *ap96*, *ap97*, *ap98*, *ap99*, *ap100*, *ap101*, *ap102*, *ap103*, *ap104*, *ap105*, *ap106*, *ap107*, *ap108*, *ap109*, *ap110*, *ap111*, *ap112*, *ap113*, *ap114*, *ap115*, *ap116*, *ap117*, *ap118*, *ap119*, *ap120*, *ap121*, *ap122*, *ap123*, *ap124*, *ap125*, *ap126*, *ap127*, *ap128*, *ap129*, *ap130*, *ap131*, *ap132*, *ap133*, *ap134*, *ap135*, *ap136*, *ap137*, *ap138*, *ap139*, *ap140*, *ap141*, *ap142*, *ap143*, *ap144*, *ap145*, *ap146*, *ap147*, *ap148*, *ap149*, *ap150*, *ap151*, *ap152*, *ap153*, *ap154*, *ap155*, *ap156*, *ap157*, *ap158*, *ap159*, *ap160*, *ap161*, *ap162*, *ap163*, *ap164*, *ap165*, *ap166*, *ap167*, *ap168*, *ap169*, *ap170*, *ap171*, *ap172*, *ap173*, *ap174*, *ap175*, *ap176*, *ap177*, *ap178*, *ap179*, *ap180*, *ap181*, *ap182*, *ap183*, *ap184*, *ap185*, *ap186*, *ap187*, *ap188*, *ap189*, *ap190*, *ap191*, *ap192*, *ap193*, *ap194*, *ap195*, *ap196*, *ap197*, *ap198*, *ap199*, *ap200*, *ap201*, *ap202*, *ap203*, *ap204*, *ap205*, *ap206*, *ap207*, *ap208*, *ap209*, *ap210*, *ap211*, *ap212*, *ap213*, *ap214*, *ap215*, *ap216*, *ap217*, *ap218*, *ap219*, *ap220*, *ap221*, *ap222*, *ap223*, *ap224*, *ap225*, *ap226*, *ap227*, *ap228*, *ap229*, *ap230*, *ap231*, *ap232*, *ap233*, *ap234*, *ap235*, *ap236*, *ap237*, *ap238*, *ap239*, *ap240*, *ap241*, *ap242*, *ap243*, *ap244*, *ap245*, *ap246*, *ap247*, *ap248*, *ap249*, *ap250*, *ap251*, *ap252*, *ap253*, *ap254*, *ap255*, *ap256*, *ap257*, *ap258*, *ap259*, *ap260*, *ap261*, *ap262*, *ap263*, *ap264*, *ap265*, *ap266*, *ap267*, *ap268*, *ap269*, *ap270*, *ap271*, *ap272*, *ap273*, *ap274*, *ap275*, *ap276*, *ap277*, *ap278*, *ap279*, *ap280*, *ap281*, *ap282*, *ap283*, *ap284*, *ap285*, *ap286*, *ap287*, *ap288*, *ap289*, *ap290*, *ap291*, *ap292*, *ap293*, *ap294*, *ap295*, *ap296*, *ap297*, *ap298*, *ap299*, *ap300*, *ap301*, *ap302*, *ap303*, *ap304*, *ap305*, *ap306*, *ap307*, *ap308*, *ap309*, *ap310*, *ap311*, *ap312*, *ap313*, *ap314*, *ap315*, *ap316*, *ap317*, *ap318*, *ap319*, *ap320*, *ap321*, *ap322*, *ap323*, *ap324*, *ap325*, *ap326*, *ap327*, *ap328*, *ap329*, *ap330*, *ap331*, *ap332*, *ap333*, *ap334*, *ap335*, *ap336*, *ap337*, *ap338*, *ap339*, *ap340*, *ap341*, *ap342*, *ap343*, *ap344*, *ap345*, *ap346*, *ap347*, *ap348*, *ap349*, *ap350*, *ap351*, *ap352*, *ap353*, *ap354*, *ap355*, *ap356*, *ap357*, *ap358*, *ap359*, *ap360*, *ap361*, *ap362*, *ap363*, *ap364*, *ap365*, *ap366*, *ap367*, *ap368*, *ap369*, *ap370*, *ap371*, *ap372*, *ap373*, *ap374*, *ap375*, *ap376*, *ap377*, *ap378*, *ap379*, *ap380*, *ap381*, *ap382*, *ap383*, *ap384*, *ap385*, *ap386*, *ap387*, *ap388*, *ap389*, *ap390*, *ap391*, *ap392*, *ap393*, *ap394*, *ap395*, *ap396*, *ap397*, *ap398*, *ap399*, *ap400*, *ap401*, *ap402*, *ap403*, *ap404*, *ap405*, *ap406*, *ap407*, *ap408*, *ap409*, *ap410*, *ap411*, *ap412*, *ap413*, *ap414*, *ap415*, *ap416*, *ap417*, *ap418</*

^aFinnish Museum of Natural History, University of Helsinki, Finland; ^bOEB, Faculty of Biology and Environmental Sciences, University of Helsinki, Finland; ^cECOFON, Faculty of Biology and Environmental Sciences, University of Helsinki, Finland; ^dNew York Botanical Garden, USA; ^eUniversity of Maryland USA; ^fRoyal Botanic Garden, Edinburgh, UK; ^gInstitute of Biotechnology, University of Helsinki, Finland.
*Email: jaakko.hyvonen@helsinki.fi

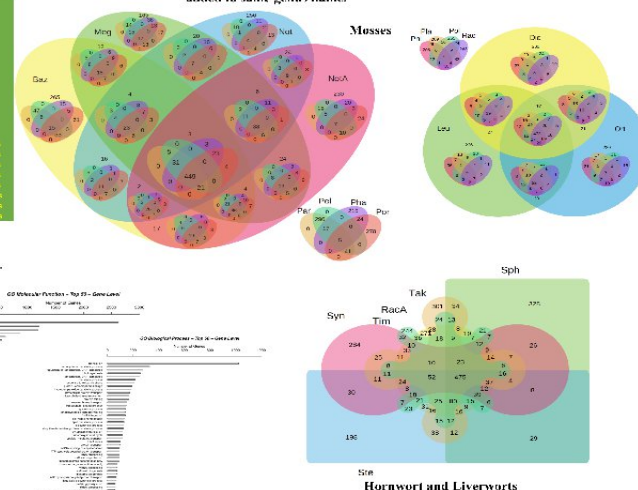
Transcriptomics plays a central role in understanding the genetic underpinning of post-transcriptional regulation. In recent years, transcriptomics has not only served as a model for unravelling post-transcriptional regulatory events but has also provided a basis for species-level comparative transcriptomics. In the absence of genomic data, transcriptomics not only allows for *de novo* phylo-transcriptomics approaches but also enables the detection of specific genes, which are specific to certain clades of organisms. We present the first reference genophenotypic transcriptome of *Alnus* species based on a large number of transcriptomic data sets. The use of transcriptomic genophenotypic transcriptomics is of an early embryonic type. We also highlight the role of transcriptomics in identifying early genes and possible phylogenetic pressures on ancestral gene evolution across land plants.

[illegible]

Transcriptome Assembled transcripts	130871
Number of Supert transcripts	80880
Percent GC	46.55
*Transcriptome statistics based on ALL transcript contigs	
Contig N50:	3191bp
Median contig length	653bp
Average contig	1533.25bp
*Transcriptome statistics based on longest isoform	
Contig N50	2334bp
Median contig length	3949bp
Average contig	921.372bp
Number of coding regions predicted using GeneMark-ST (Stap)	40812
Number of coding regions predicted using GeneMark-ES (Stap)	24336
Number of transcripts associated via nVINO	39499
Number of superttranscripts associated via nVINO	11696
Number of transcripts mapped to KEGG	22996
Number of superttranscripts mapped to KEGG	2199

***Blasia pusilla* gametophyte transcriptome summary statistics**

Yeast diagram showing the distribution of the shared *Blasia pusilla* transcripts across the hornwort and liverwort LCP gametophyte transcriptome. * The first three letter defines the species in the table with (A) added to same genus name.



Gene ontology classification at cellular, biological and molecular functions.

Acknowledgements Funding of F and for this project 795597, 791163, and to CSC-IT Centre for Sustainable IT provided.

End to End Analysis and Machine learning

Entire metagenomics analysis: <https://codeberg.org/gsablok/microbiomemag>

Regression Modelling: <https://codeberg.org/gsablok/biomereg>

Classification Modelling: <https://codeberg.org/gsablok/biomeboost>

LSTM: <https://codeberg.org/gsablok/microbiomeRNN>

July-2011- Sept 2014: Postdoc at FEM, Italy, Established computing cluster and diverse set of analysis from transcriptomics to genomics, genome, metagenomics, bacterial and fungal genomics.

Draft Genome Sequence of the Nitrogen-Fixing *Rhizobium sulae* Type Strain IS123^T Focusing on the Key Genes for Symbiosis with its Host *Hedysarum coronarium* L.

Gaurav Sablok^{1*}, Riccardo Rossini¹, Tonien Seaman¹, Robin van Velsen¹, Elias Polone¹, Alessio Giacomini¹, Nicola La Porta^{1,2}, Rene Geurts¹, Rosella Murenu¹ and Andrea Squartini¹

SCIENTIFIC REPORTS

OPEN

SeagrassDB: An open-source transcriptomics landscape for phylogenetically profiled seagrasses and aquatic plants

Received: 12 June 2017
Accepted: 11 December 2017
Published online: 09 February 2018

Gaurav Sablok¹, Regan J. Hayward², Peter A. Davey¹, Rosiane P. Santos¹, Martin Schliep¹, Anthony Larkum¹, Mathieu Pernice¹, Rudy Dolferus¹ & Peter J. Ralph²

Environment, Development and Sustainability
<https://doi.org/10.1007/s10668-023-03144-w>

Metabarcoding analysis of the bacterial and fungal communities during the maturation of preparation 500, used in biodynamic agriculture, suggests a rational link between horn and manure

Marina Zanardo¹, Matteo Giannattasio², Gaurav Sablok³, Massimo Pindo³, Nicola La Porta^{3,4}, Michele Lorenzetti⁵, Carlo Noro⁵, Piergiorgio Stevanato¹, Giuseppe Concheri¹, Andrea Squartini¹

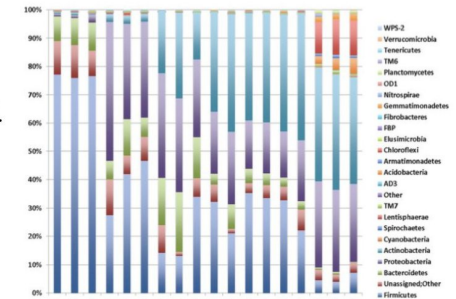
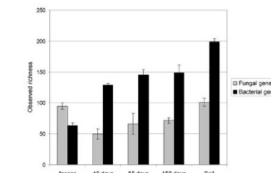
Received: 3 February 2022 / Accepted: 10 March 2023
© The Author(s), under exclusive licence to Springer Nature B.V. 2023

Sept 2014-Sept 2016: Research Fellow, University of Technology Sydney, Australia.

1. Established Computing Cluster for the C3.
2. Developed the analysis for the transcriptomics and other omics analysis.
3. Developed and conducted all the transcriptomics analysis for the SeagrassDB and a collaborator build the DB.

TABLE 1 | Comparative genome features of *Rhizobium sulae* IS123^T, its co-specific *R. sulae* WSM1592, and its two taxonomically closest relatives *R. leguminosarum* biovar *viciae* 3841 and *R. etli* CFN42.

Features	<i>R. sulae</i> IS123 ^T	<i>R. sulae</i> WSM1592	<i>R. leguminosarum</i> bv. <i>viciae</i> 3841	<i>R. etli</i> CFN42
Genome (bp)	7,889,576	7,530,820	7,751,309	6,530,228
GC average %	59.88%	59.87	60.86%	60.54
rRNA operons	6	G5	3	3
tRNA	51	G47	52	50



PhyloEvolve: A scalable toolkit for curation, manipulation, and visualization of gene and marker alignments

PHYLOEVOLVE

phylogenomics made easy for gene and marker alignment

Author Gaurav Sablok,

Email: gsablok@proton.me

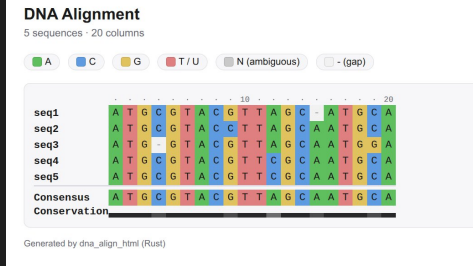
Usage: phyloEvolve <COMMAND>

Commands:

alignmerge	merge all the alignment into a single string
alignmergeinterval	alignmerge the specific region of the alignment
same-alignment	remove same alignment
alignmentstats	alignmentstats
filter-site	filter the sites with the given base
filter-all	removes same bases across all the alignment
filter-block	collineratiy block based alignment filtering
alignment-view	allows for the view of the alignment
alignment-clipview	allows for the embedded view of the clipped Alignment
sitereplace	replace the specific sites based on the site probability
protein-stat	estimate the protein stats
indelreplace	indel substituter
motif-search	search for the specific protein and the nucleotide motifs
up-down-stream	specific part of the alignment and the upstream and the downstream of alignment
site-alignment	extract the places of the specified base for LD analysis
plotter	frequency plot for the alignments
nucleotidecolour	color coded map
proteomecolour	proteome heatmap
alignment-html	alignment to html
sub-alignment-html	sub alignment to html
alignment-convert	convert alignment
padded-sub-alignment-html	padded sub alignment to html
help	Print this message or the help of the given subcommand(s)

Options:

-h, --help	Print help
-V, --version	Print version



How My Expertise and Research Plan Will Deliver the Postdoctoral Project

Why i find your job offer, opportunity, chance and position interesting:

- I have worked extensively in the areas of bioinformatics, data analysis, application and software development. This position offers me a opportunity to integrate all of this, develop it further, and to develop new areas of the analysis and integration for all species of interest.

Why i should be considered and employed:

- **Single and Lead Bioinformatician in all the jobs and places of employment.**
- **Lead all the analysis and manuscript preparation and writing (Authors contribution section).**
- **Developed and Lead all the software development, computing cluster management.**
- I have wide proficiency in data analysis and establishing new approaches for these technologies and have worked across microbial and plant bioinformatics.
- Have wide proficiency in all types of data analysis bioinformatics, software development and machine and deep learning.
- Extremely open minded, loyal and collaborative person at the personal and position level.
- Collaborative, like working with people who talk about solution by bringing all together.
- Solution and result oriented.
- Having wide proficiency across several aspects of bioinformatics such as genome assembly, transcriptomics, annotation, phylogenomics, machine learning and evolutionary aspects incorporating different types of short read and long read sequencing.



Group/Lab Integration and Development: Supporting specific groups activities, project for ongoing projects and to address and cater the specific needs. Develop with the Group head, new approaches, cross group or lab collaboration and seek the holistic growth and approach.